

On the Spectral Decomposition of the Collatz Iteration Count

Localization of the 3-Expansion Difficulty

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Abstract: We decompose the Fourier spectrum of the Collatz iteration count $\sigma(n)$ into dyadic discrete peaks and a broadband continuous residual, proving rigorously that the residual is exclusively carried by the odd-step recursion (the 3-expansion term), accounting for 90.5% of the total energy. Via the $q=4$ spectral-trough amplitude decomposition, we identify the “non-closing odd class” S_3 as the concrete algebraic source of this residual. Furthermore, we refute the common intuition that the 625-lattice ladder originates from the $+v_2(n)$ linear term; instead, it arises from odd-core imprint geometry. This work provides the first precise spectral localization of the “ $3n+1$ step” as the core carrier of Collatz difficulty.

1. Introduction

Let $\sigma(n)$ denote the total stopping time of n . While previous studies focused on mean values and distributions (Tao 2019, Terras 1976), our large- N ($N=20000$) DFT analysis reveals a clear spectral dichotomy: **dyadic lattice lines** (multiples of $k=625$) coexist with a **broadband continuous residual** (90.5% of energy), the latter corresponding exactly to the 3-expansion term in the odd-step recursion $f(2m-1)=1+f(3m-1)$.

2. Main Theorem: Difficulty Decomposition (S-F3)

Theorem 1 (Difficulty Decomposition) For a window of size $N=2^K \cdot 5^2$, the spectral energy of σ satisfies:

$$E_{\text{total}} = E_{\text{dyadic}} (9.5\%) + E_{\text{broadband}} (90.5\%)$$

The dyadic peaks ($f = j/80$) arise from the even-step recursion $\sigma(2m)=1+\sigma(m)$, a 2-regular structure. The broadband residual, rather than vanishing under 3-adic binning, exhibits increasing class-mean ranges with modulus 3^j (slope $+0.886$), indicating non-3-adic continuity (S-J2) and genuine complexity.

3. Spectral-Trough Amplitude Closure (S-F4)

For the $q=4$ spectral trough ($k = N/4$), we derive the exact decomposition:

$$X[N/4] = [\Sigma \sigma(3m-2) + 3q - 3 - S_3] + i[\Sigma \sigma(m) - \Sigma \sigma(3m-1) - q + 3]$$

where $S_3 = \Sigma \sigma(4m-1)$ is the “non-closing odd class” ($4m-1 \rightarrow 6m-1 \rightarrow 9m-1 \rightarrow \dots$ an infinite odd chain). S_3 contributes $\sim 8.9\times$ the magnitude of the closed-form part, directly quantifying the 3-expansion's dominance over the spectral trough.

4. Ladder Residence Refutation (S-K2)

The hypothesis that the 625-lattice ladder stems from the $+v_2(n)$ additive term is refuted. After odd-core stripping ($\sigma(n)=\sigma(\text{core}(n))+v_2(n)$), the 4 discrete peaks *increase* in proportion ($0.0952 \rightarrow 0.1103$), and the odd-core signal $\sigma(\text{odd } n)$ itself already exhibits the ladder (proportion 0.0705). Conclusion: **The ladder originates from odd-core imprint geometry (dyadic orbit alignment), not from the v_2 linear term.**

Keywords: Spectral decomposition · 3-expansion · odd-core imprint · continuous residual · S-F3/S-F4/S-K2